

Morphic Studio AI Intelligence & Knowledge Architecture

Document 74 — AI Planning, Reasoning Pipelines & Autonomous Task Decomposition Specification

Version: 1.0

Status: Approved

1. Purpose

This document defines the AI Planning, Reasoning Pipelines, and Autonomous Task Decomposition Framework for Morphic Studio.

Its purpose is to enable AI agents to transform high-level creative objectives into structured, explainable, and adaptable execution plans while preserving human oversight, creative intent, and operational transparency.

Planning serves as the executive intelligence layer connecting user goals to coordinated agent execution.

2. Vision

Planning should resemble the work of an experienced creative producer.

Rather than immediately generating outputs, AI first understands objectives, evaluates constraints, identifies dependencies, constructs an execution strategy, coordinates specialists, monitors progress, and continuously adapts when conditions change.

Planning is intentional rather than reactive.

3. Design Principles

Planning follows these principles:

- Goals before actions.

- Plans remain explainable.
 - Human approval governs significant decisions.
 - Adaptation is continuous.
 - Dependencies are explicit.
 - Failures trigger replanning rather than abandonment.
 - Creative flexibility coexists with structured execution.
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4. Planning Hierarchy

Planning occurs across multiple levels:

Strategic Planning

- Long-term project goals
- Publication objectives
- Production strategy

Tactical Planning

- Milestones
- Deliverables
- Workflow stages

Operational Planning

- Individual tasks
- Agent assignments
- Immediate execution

Micro Planning

- Tool usage
- API calls
- Local reasoning steps
- Validation activities

Each planning level inherits constraints from higher levels.

5. Goal Analysis

Incoming requests are analyzed to determine:

- Desired outcomes
- Success criteria

- Creative constraints
- Technical constraints
- Available resources
- Required approvals
- Estimated complexity
- Priority

Goal analysis precedes execution.

6. Task Decomposition

Complex objectives are divided into manageable tasks.

Example:

Create Graphic Novel

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Story Planning

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Character Development

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World Building

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Storyboard Generation

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Illustration

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Editing

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Publishing

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Marketing Assets

Each task may be further decomposed until executable.

7. Dependency Graph

Planning identifies dependencies including:

- Sequential requirements
- Parallel opportunities
- Resource conflicts
- Canon dependencies
- Approval gates
- Asset prerequisites
- Timeline constraints

Dependencies are represented explicitly to support reliable orchestration.

8. Reasoning Pipelines

Reasoning progresses through stages:

Understand Request

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Retrieve Context

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Analyze Constraints

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Construct Plan

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Validate Plan

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Assign Agents

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Execute

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Review Results

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Refine if Needed

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Deliver Output

Every stage is observable for auditing.

9. Agent Coordination During Planning

Planning coordinates specialist agents including:

- Story Architect
- Character Coach
- World Builder
- Canon Guardian
- Art Director
- Producer
- Workflow Manager
- Research Assistant

The orchestrator assigns work according to expertise and workload.

10. Creative vs Deterministic Planning

The framework distinguishes between:

Creative Planning

- Brainstorming
- Narrative exploration
- Artistic direction
- Alternative concepts

Deterministic Planning

- Publishing workflows
- Asset processing
- Scheduling
- Validation
- Compliance

Different planning strategies are applied according to task type.

11. Human Approval

Approval checkpoints may occur before:

- Major project creation
- Canon modifications
- Publishing
- Large AI expenditures
- Destructive actions
- Organizational policy changes

Users determine approval thresholds.

12. Iterative Refinement

Plans remain adaptable.

The system may:

- Add missing tasks
- Remove unnecessary work
- Reorder execution
- Reassign agents
- Improve estimates
- Incorporate user feedback
- Respond to changing requirements

Planning is continuously refined throughout execution.

13. Failure Recovery & Replanning

When execution fails, the framework may:

- Retry tasks
- Select alternative agents
- Modify execution order
- Request additional context
- Escalate to the user
- Produce a revised plan

Failures are treated as planning inputs rather than terminal events.

14. Explainable Planning

Users may inspect:

- Overall objectives
- Task hierarchy
- Agent assignments
- Dependencies
- Decision rationale
- Estimated timelines
- Approval points
- Current execution status

Planning transparency strengthens user trust.

15. Performance Optimization

Planning optimizes for:

- Reduced execution time
- Efficient agent utilization
- Parallel processing
- Minimal redundant reasoning
- Balanced resource consumption
- Context reuse
- Scalable orchestration

Optimization never overrides correctness or creator intent.

16. Integration

The Planning Framework integrates with:

- Multi-Agent Orchestrator
- Memory Framework
- Knowledge Graph
- Workflow Engine
- Story Workspace
- Character Studio
- World Builder
- Production Studio
- Asset Manager
- Publishing Center
- Analytics
- Administration

Planning becomes the executive control layer across the entire platform.

17. Acceptance Criteria

The Planning Framework is successful if:

- High-level goals are decomposed into executable tasks.
 - Dependencies are identified and managed explicitly.
 - Agent coordination remains efficient and transparent.
 - Human approval governs meaningful creative decisions.
 - Plans adapt to changing conditions without losing coherence.
 - Failures trigger intelligent replanning.
 - Users can inspect and understand planning decisions.
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Conclusion

The Morpheic Studio AI Planning, Reasoning Pipelines, and Autonomous Task Decomposition Framework establishes the executive reasoning capability of the platform. Through hierarchical planning, explainable reasoning, structured task decomposition, dependency management, adaptive replanning, and coordinated multi-agent execution, the framework enables AI to transform ambitious creative goals into reliable, transparent, and creator-controlled workflows that scale from individual scenes to enterprise-scale creative productions.